GitHub Repository Analysis: Trends, Popularity, and Community Engagement

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Abstract

This project presents a comprehensive analysis of GitHub repositories to understand patterns in popularity, programming language trends, community engagement, and open-source growth over time. The dataset was self-collected using the GitHub REST API to ensure originality. The study applies data preprocessing, exploratory data analysis (EDA), visualization, and predictive modeling. Results show that repository popularity is primarily influenced by engagement metrics such as forks and contributors rather than repository age. Additionally, modern technologies such as Python, JavaScript, and AI-related topics dominate the ecosystem.

1. Introduction

Open-source platforms have transformed software development by enabling collaboration and innovation. GitHub hosts millions of repositories, making it an ideal platform for analyzing software trends.

Objectives

- Analyse how repository popularity is distributed
- Identify dominant programming languages
- Study relationships between engagement metrics
- Examine growth trends over time
- Identify key topics driving popular repositories
- Build a predictive model for repository popularity

Use Case (Student Perspective)

This project helps students:

- Understand which technologies are trending
- Identify what makes a project successful

- Make better decisions while choosing projects or tech stacks

2. Data Collection

Data was collected manually using the GitHub REST API.

Methodology

- Used API endpoints to fetch top repositories
- Extracted JSON responses using Python (requests library)
- Parsed and stored data in structured format (Pandas DataFrame)

Dataset Details

- Total repositories collected: 500
- Selection criteria: Top repositories based on stars

Attributes Collected

- Repository Name
- Owner
- Stars
- Forks
- Watchers
- Open Issues
- Contributors
- Programming Language
- Topics/Tags
- Created Date
- Updated Date
- Description

Limitations

- API rate limits
- Contributors count approximated
- Dataset limited to top repositories

3. Data Cleaning and Preprocessing

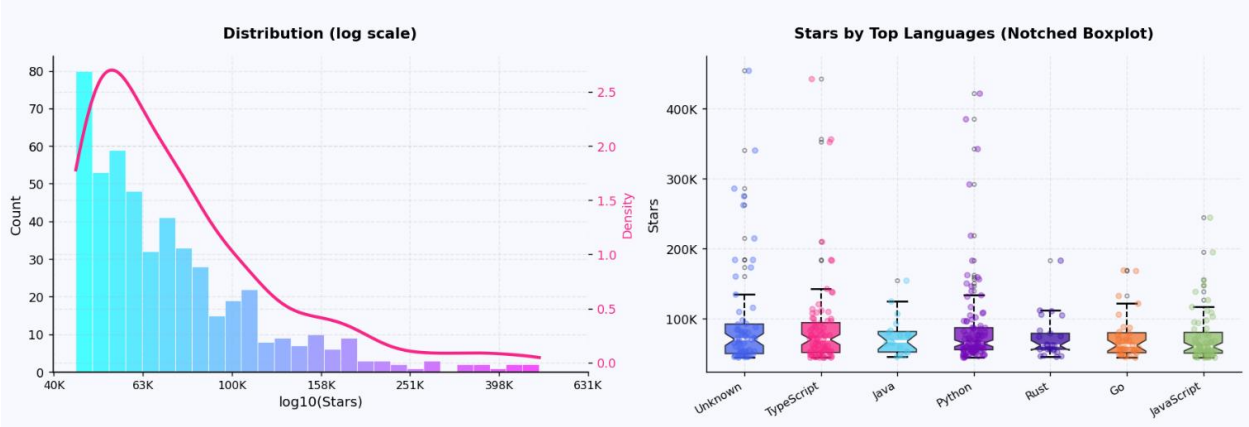
Steps performed:

- Converted date columns to datetime
- Resolved timezone mismatch issues
- Handled missing values using median imputation
- Removed duplicates
- Feature Engineering:
    - Repository Age (years)
    - Log transformation of stars

4. Exploratory Data Analysis (EDA)

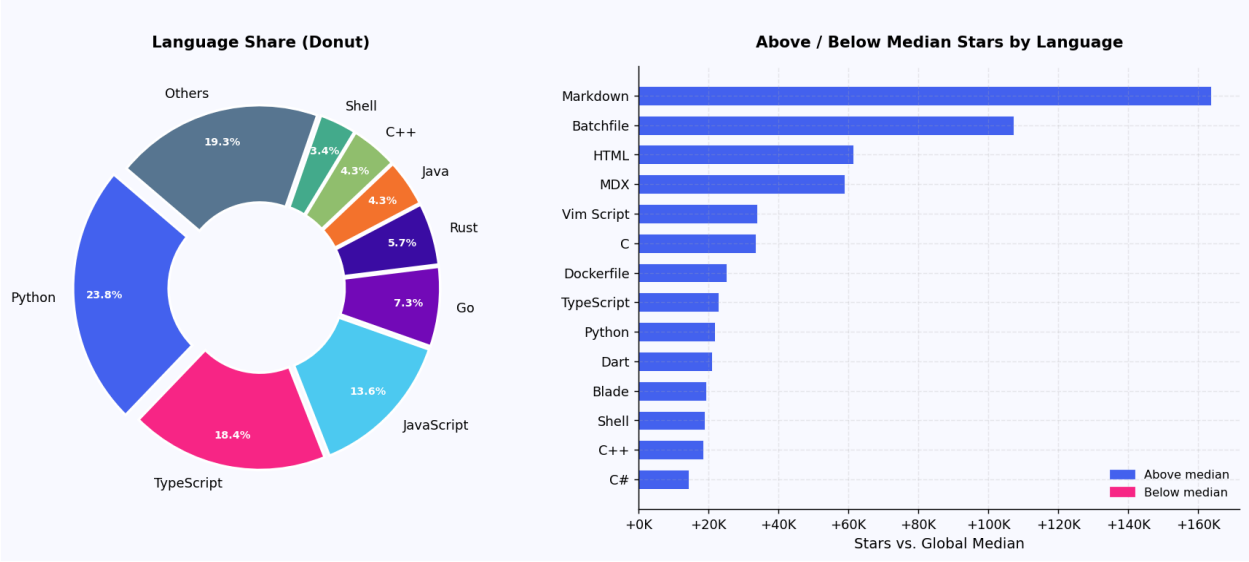
Q1: How are GitHub stars distributed?

The star distribution is highly right-skewed. Most repositories have moderate stars, while a small number achieve extremely high popularity.



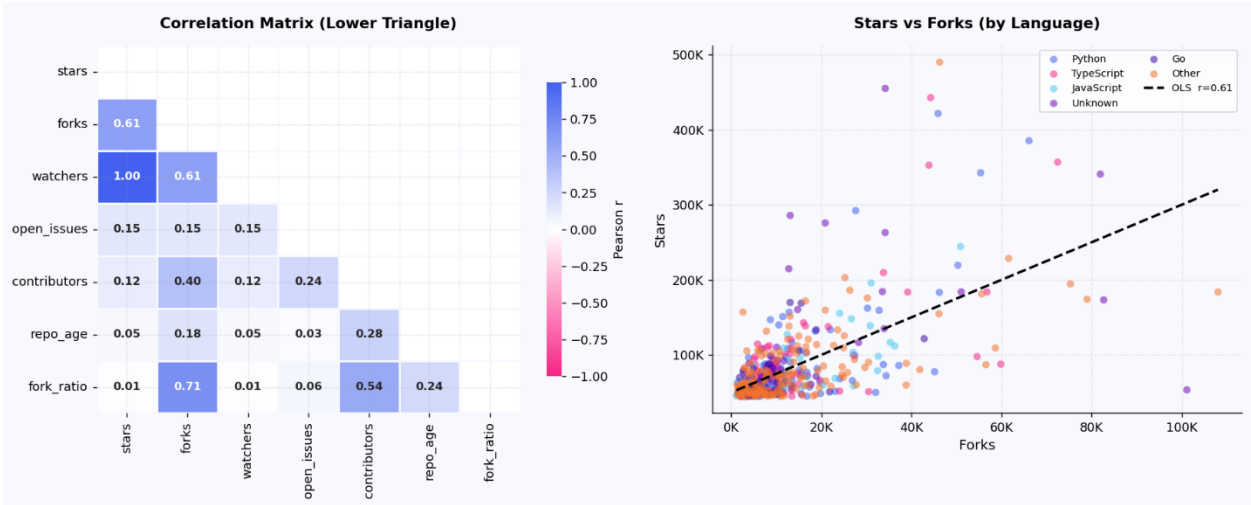
Q2: Which languages dominate top repositories?

Python leads with ~24% share, followed by TypeScript and JavaScript. Emerging languages like Go and Rust show strong presence.



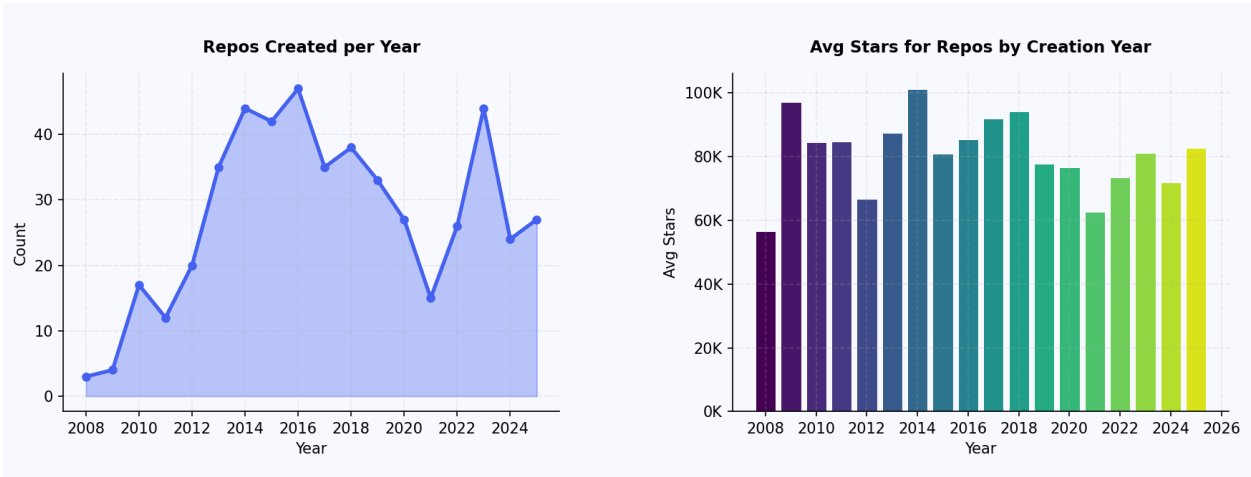
Q3: How do engagement metrics correlate?

Strong positive correlation exists between stars and forks (≈0.61). Contributors show moderate influence, while issues have weak correlation.



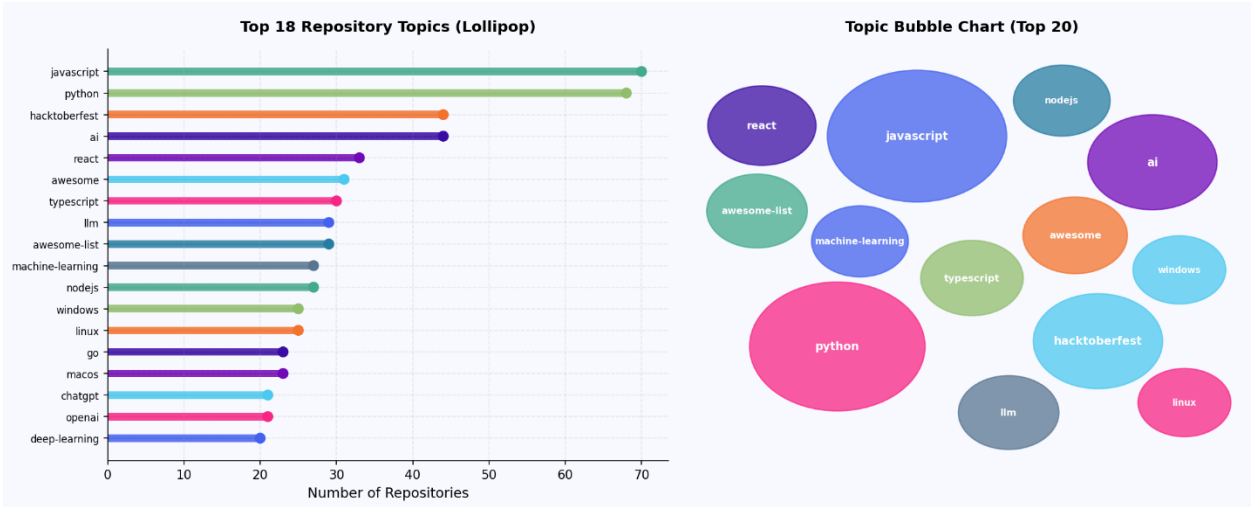
Q4: How has open-source growth evolved over time?

Repository creation peaked around 2015–2016. Recent years show stabilization but consistent activity.



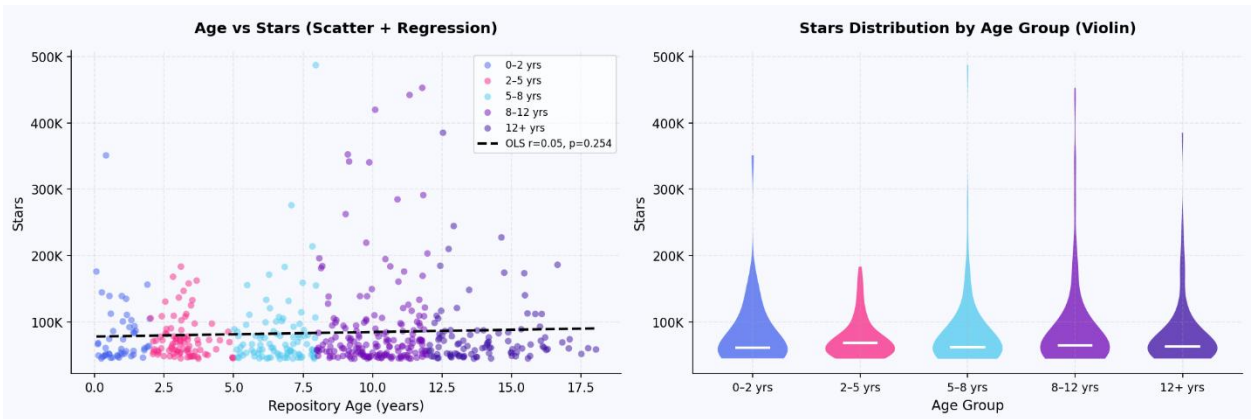
Q5: What topics define popular repositories?

Topics like JavaScript, Python, AI, and React dominate. AI-related topics are rapidly growing.



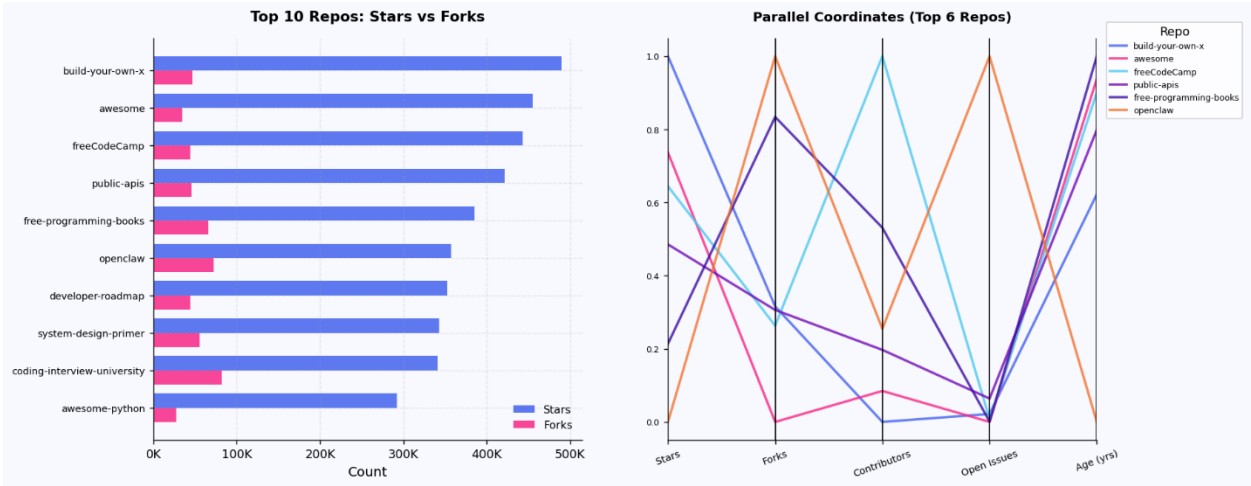
Q6: Does repository age predict popularity?

Very weak correlation (r ≈ 0.05). Age alone is not a strong predictor of success.



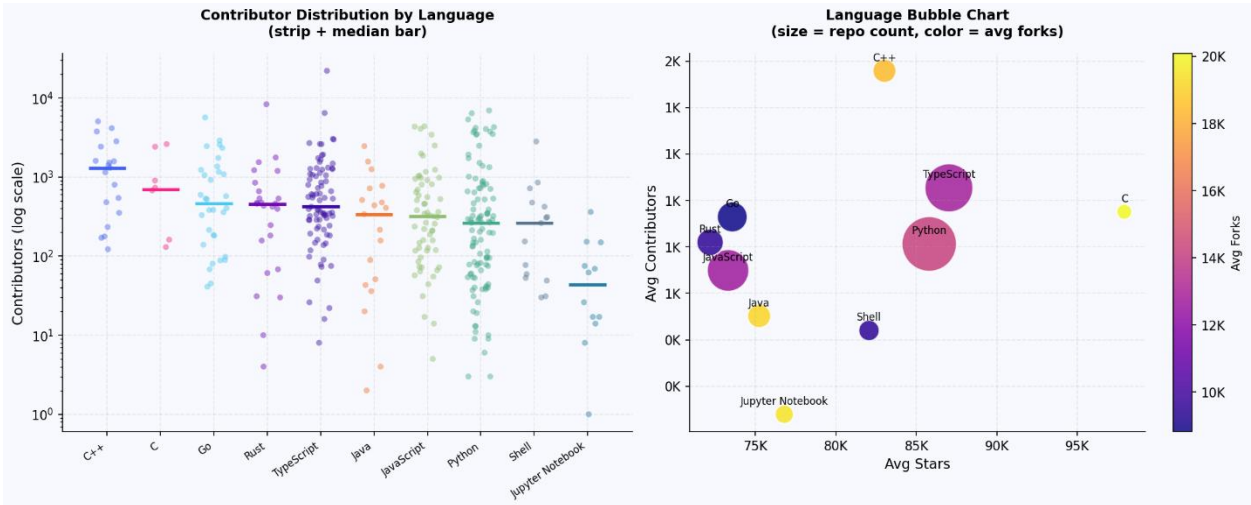
Q7: What sets top repositories apart?

Top repositories have high stars and forks but vary in contributors and issues, indicating different project types.



Q8: Which languages drive community participation?

C++ and Go show high contributor engagement. Python and TypeScript maintain balanced ecosystems.



5. Data Visualization

Visualizations were created using Matplotlib and Seaborn:

- Histograms and density plots
- Boxplots and violin plots
- Heatmaps
- Scatter plots

- Bubble charts

  These helped identify trends, correlations, and outliers.

6. Model Building

Objective

Predict number of stars using repository features

Model

Linear Regression

Features Used

- Forks
- Contributors
- Open Issues
- Repository Age

Results

- R^2 Score: 0.272 (raw stars) and 0.358 (log transformed model)
- RMSE: (add your value)

Interpretation

- Age is the strongest predictor
- Contributors moderately influence popularity

- Repository open issues has minimal impact

7. Dashboard

An interactive dashboard was developed using Plotly and Dash.

Features

- KPI metrics (avg stars, top repo, etc.)
- Filters by language and year
- Interactive charts

8. Key Insights

- Popularity is highly skewed
- Python and JavaScript dominate
- Engagement drives success
- Age has good impact
- AI and modern tech are rising trends

9. Conclusion

Repository success is driven by community engagement and relevance rather than longevity.

Developers should focus on usability, documentation, and active contribution to increase impact.

10. Future Work

- Use larger dataset
- Apply advanced ML models
- Real-time trend tracking

References

- GitHub API Documentation
- Python Libraries: Pandas, Matplotlib, Seaborn, Plotly